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BSECE – 1B

Appendix A.

1. **Original Records:** [108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119]
2. **Rotated Records:** [109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 108]
3. **Target Record:** 113
4. **Position:** Index 4
5. **Number of Comparisons:**
   - Linear Search: 5
   - Rotated Binary Search: 4
6. **Comparison**: Rotated Binary Search is more efficient than Linear Search because it required fewer comparisons (4 vs. 5).

Appendix B.

Department A dataset: [108, 115, 122, 129, 136, 143, 150, 157, 164, 171]
Department B dataset: [108, 115, 122, 175, 180, 185, 190, 195, 200, 205, 210]
Consolidated dataset: [108, 115, 122, 129, 136, 143, 150, 157, 164, 171, 175, 180, 185, 190, 195, 200, 205, 210]
Number of records before and after consolidation
Before: 21
After: 18
Comparison count: 3 (Shared records: 108, 115, 122)
Ordering verification: Verified. The consolidated dataset is in ascending order.
Time and space complexity
Time Complexity: O(n + m)
Space Complexity: O(n + m)
Final Output
Shared Records: [108, 115, 122]
Total Records After Consolidation: 18

Appendix C.

Generated order dataset: [213, 226, 239, 252, 265, 278, 291, 304, 317, 330, 343, 356, 369, 382, 395, 408, 421, 434]
Sorted, reverse, and random inputs
Sorted Input: [213, 226, 239, 252, 265, 278, 291, 304, 317, 330, 343, 356, 369, 382, 395, 408, 421, 434]
Reverse Input: N/A
Random Input: N/A
Comparison counts
Bubble Sort: 153
Selection Sort: 153
Insertion Sort: 17

Movement counts

Bubble Sort Swaps: 0

Selection Sort Swaps: 0

Insertion Sort Shifts: 0

Execution times

Bubble Sort: 153 comparisons, 0 swaps

Selection Sort: 153 comparisons, 0 swaps

Insertion Sort: 17 comparisons, 0 shifts

Sorting verification: Verified. All sorting algorithms produced the same ascending order.

Big O analysis

Bubble Sort: $O(n^2)$

Selection Sort: $O(n^2)$

Insertion Sort: Best: $O(n)$, Average/Worst: $O(n^2)$

Best-, average-, and worst-case observations

Bubble Sort: Best $O(n)$, Average $O(n^2)$, Worst $O(n^2)$

Selection Sort: Best $O(n^2)$, Average $O(n^2)$, Worst $O(n^2)$

Insertion Sort: Best $O(n)$, Average $O(n^2)$, Worst $O(n^2)$

Final Output

Sorted Orders: [213, 226, 239, 252, 265, 278, 291, 304, 317, 330, 343, 356, 369, 382, 395, 408, 421, 434]

Most Efficient for this Dataset: Insertion Sort (17 comparisons, 0 shifts)